

# Juan José Aguado

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## EDUCATION

### Pontificia Universidad Javeriana

*Systems and Computer Engineering (Computer Science)*

Cali, VA

Aug. 2020 – Jun. 2025

## EXPERIENCE

### Tasker - Generative AI Models

Jun. 2024 – Present

*Outlier*

*Remote*

- Evaluated the performance of generative AI models (text, audio, image) based on criteria such as precision, instruction-following, and safety.
- Conducted evaluations in English, ensuring a high standard of language proficiency and communication in reports.
- Provided feedback on model's outputs, helping improve their accuracy and compliance with security protocols.

### Database Administrator

Jan. 2024 – Jun. 2024

*Fundación Casa de Colombia*

*Cali, VA*

- Spearheaded the design, creation, and deployment of a comprehensive database for a software solution aimed at streamlining operations at Casa de Colombia.
- Developed and maintained PostgreSQL databases to efficiently manage and store crucial information, supporting the provision of essential health and education services to children.
- Utilized Jira for project management and issue tracking, ensuring efficient workflow and task management.
- Employed Git and Bitbucket for version control and collaborative development, maintaining a robust and organized codebase.

### Research Assistant - Applied Software Engineering Research Group

Jun. 2024 – Present

*Pontificia Universidad Javeriana*

*Cali, Colombia*

- Working on the university's automatic email sending service. Developed and modified functionalities using Java and Maven.
- Utilized Postman to test and validate API endpoints for automated email delivery.
- Collaborated with teams to enhance email service performance and maintainable code structure.
- Integrated solutions and performed troubleshooting on email service errors and logs to improve user experience.

### Data Scientist

Oct. 2024 – Present

*Atinna*

*Cali, Colombia*

- Developing social listening systems by consuming and transforming large-scale datasets using Python, Git, Hugging Face, OpenAI, and LangChain for LLM-based processing and analysis.
- Utilizing embeddings for clustering and semantic similarity tasks to identify meaningful relationships and topic structures within multilingual social media data.
- Performing fine-tuning experiments through few-shot learning and DSPy, enhancing LLM performance on sentiment, stance, and political alignment classification tasks.
- Designing and managing a relational database with Azure SQL to store structured listening outputs and metadata for dynamic Power BI visualizations.
- Integrating Langfuse with the backend to monitor and version prompts directly from the Langfuse platform, ensuring traceability and controlled prompt evolution.
- Building a backend architecture to automate the complete workflow of social listening pipelines, including ingestion, model execution, and report generation.
- Creating interactive visualizations, dynamic maps, and strategic analytics dashboards to support data-driven decisions and client reporting.
- Optimizing collaborative Git workflows to enhance productivity, maintain reproducibility, and improve model deployment cycles.

## PROJECTS

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### UVa Programming Problems | *Python*

January 2023 – June 2023

- Solved numerous programming problems from the UVa Online Judge, a platform hosted by the University of Valladolid with over 4300 problems.
- Utilized various data structures such as arrays, linked lists, trees, queues, and graphs to optimize solutions.
- Implemented algorithms including bisection, binary search, backtracking, dynamic programming, and greedy algorithms to solve complex problems.
- Gained deep understanding of problem-solving techniques and algorithm optimization through continuous practice and refinement of solutions.

### Statistical Analysis and Data Science Projects | *R, Markdown*

January 2024

- Conducted comprehensive analysis of the real estate market in Cali, Colombia, using data from B&C real estate agency, focusing on property prices, locations, features, and sales.
- Preprocessed and cleaned real estate data, analyzed descriptive statistics, and created visualizations to communicate findings effectively.
- Performed various statistical simulations, including Monte Carlo Simulation to estimate the value of  $\pi$ , and analyzed bias, efficiency, and consistency of different estimators.
- Verified the Central Limit Theorem through simulations, observed its implications on sample distributions, and constructed confidence intervals using bootstrap methods.
- Built multiple linear regression models to predict real estate prices based on the area, validated model assumptions, performed hypothesis testing, and compared models to find the best fit.
- Developed a deep understanding of statistical analysis and data science techniques using R, enhancing skills in data manipulation, visualization, and modeling.

## COURSES

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### HarvardX CS50 | *Django, HTML, Git, CSS, Javascript, SQL*

April 2024

- Developed a full-stack web application using Django, implementing both the backend with RESTful APIs and the frontend with HTML, CSS, and JavaScript.
- Utilized Git and GitHub for version control and collaboration, ensuring efficient project management and code integrity.
- Implemented interactive features and dynamic content using JavaScript for seamless user experience.
- Designed and managed relational databases with SQL, ensuring data integrity and efficient query handling.
- Applied best practices in web security, testing, and deployment, ensuring robust and reliable applications.

## SKILLS

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**Programming languages:** Python, C/C++, SQL (Postgres, Azure SQL), JavaScript, HTML/CSS, R

**Frameworks:** Django

**Libraries:** pandas, NumPy, Matplotlib, Seaborn, SciKitLearn, Hugging Face, OpenAI, LangChain, DSPy, Plotly

**Developer Tools:** Git, Docker, VS Code, PostgreSQL (PgAdmin), Azure SQL Server, Jira, Bitbucket, RStudio, VirtualBox, Google Colab, Eclipse, Maven, Langfuse, Power BI

**Cloud Platforms:** Azure, AWS

**Languages:** Spanish (native), English (advanced)